Episode_1

Navigating NCBI: Retrieving and Interpreting Gene and Protein Records

Learning objectives

1. Explain what NCBI is and choose the right database (Gene, Nucleotide, Protein, Taxonomy).
2. Find a gene record by symbol and organism, avoiding the wrong-organism trap.
3. Move between Gene, Nucleotide, Protein and Taxonomy records via links.
4. Read an accession: prefix, number, version.
5. Interpret gene name, organism, location, length, CDS, protein product and function.
6. Explain how a bacterial gene (*lacZ*) and a human gene (*INS*) differ in organisation and records.
7. Repeat the workflow for any gene and organism.

Part 1. What is NCBI?

The National Center for Biotechnology Information (part of the US National Library of Medicine) is a free public library of biological data. Think of a library with a catalogue: each database is a section, and cross-references are links on the catalogue card.

Database	Holds	Answers
Gene	One page per gene: name, location, summary, links	What is this gene and where is it?
Nucleotide	DNA/RNA sequences (genomes, mRNAs)	What is the actual sequence?
Protein	Amino-acid sequences and annotation	What does the gene make?
Taxonomy	Tree of life, one ID per organism	Which organism, how classified?

GenBank vs RefSeq. GenBank is an open archive anyone can submit to (many records per gene). RefSeq is the curated, non-redundant reference set; its accessions contain an underscore (e.g. `NM_000207.3`).

Part 2. Step-by-step record retrieval

Activity 2A. Retrieve *lacZ*

1. Go to ncbi.nlm.nih.gov; set the database drop-down to **Gene**.
2. Search `lacZ`. *You should see:* a long list from many bacteria and plasmids.
3. Narrow: `lacZ AND Escherichia coli K-12 MG1655[Organism]` (or use the Species filter).
4. Click the result with description "beta-galactosidase" and Gene ID **945006**.
5. *You should see:* a page headed **lacZ** with a Summary, Genomic context and right-hand links.

Why filter? The same symbol exists in thousands of organisms. Always check the organism.

Activity 2B. Retrieve human *INS*

1. Gene database; search `INS AND Homo sapiens[Organism]`.
2. Choose symbol **INS**, name "insulin", Gene ID **3630** (ignore *INS-IGF2* and similar).

3. *You should see:* a summary on insulin and glucose regulation, band 11p15.5, and transcript/protein products.

Tip: typing a Gene ID (`945006` or `3630`) in the search bar goes straight to the record.

Part 3. Cross-referencing

The key skill is navigating, not re-searching.

3A. *INS*: Gene → Nucleotide → Protein

1. On Gene 3630, scroll to **NCBI Reference Sequences (RefSeq)**: a genomic (NC_000011.10), mRNA (NM_) and protein (NP_) table.
2. Click **NM_000207.3** (Nucleotide record).
3. In FEATURES, find `CDS` and `/protein_id="NP_000198.1"` .
4. Click the protein ID to open preproinsulin.
5. In "Related information" click **Gene** to return: a full loop.

3B. *lacZ*: Gene → Genome → Protein

1. On Gene 945006, check Genomic context and RefSeq: **NC_000913.3** and **NP_414878.1**, but no NM_ record.
2. Open NC_000913.3: the entire ~4.6 Mbp chromosome.
3. Use Find / "Highlight Sequence Features" to reach `/gene="lacZ"` , `/locus_tag="b0344"` .
4. From the CDS feature, follow the protein link to NP_414878.1.

3C. Taxonomy

Click the organism name on a Gene page. Human: Eukaryota → Metazoa → Chordata → Mammalia → Primates → *Homo sapiens* (TaxID **9606**). *E. coli*: Bacteria → Pseudomonadota → Gammaproteobacteria → Enterobacterales → Enterobacteriaceae → *Escherichia coli* (species TaxID **562**; K-12 MG1655 has its own strain ID).

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Taxonomy (organism)
     ↑
   Gene → Nucleotide (genome / mRNA) → Protein
     ↑_____________________________________|
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Part 4. Accession numbers

An accession is a permanent record ID, like an ISBN. Format: `PREFIX + NUMBER . VERSION` . Example **NM_000207.3**: `NM_` = RefSeq curated mRNA; `000207` = record number; `.3` = version.

Rule: the accession never changes; the version increases when the *sequence* changes (annotation-only edits do not). Always cite the full accession with version.

Prefix	Meaning	Example
NC_	Complete chromosome/genomic molecule	NC_000913.3, NC_000011.10
NM_	Curated mRNA	NM_000207.3
NP_	Curated protein	NP_000198.1, NP_414878.1

Prefix	Meaning	Example
NR_	Non-coding RNA	–
XM_/XP_	Computationally predicted mRNA/protein	–
NG_	RefSeqGene region	–

GenBank (non-RefSeq) accessions have no underscore, e.g. `U00096.3` (*E. coli* K-12 genome); proteins look like `AAC73446.1`.

Not accessions: Gene ID (945006, 3630), Taxonomy ID (9606), locus tag (b0344).

Part 5. Interpreting a record

Field	Tells you	***INS***	***lacZ***
Symbol / name	Standard name	INS, insulin	lacZ, beta-galactosidase
Organism	Species	*H. sapiens*	*E. coli* K-12 MG1655
Location	Where in genome	11p15.5	b0344; ~362,455–365,529 on NC_000913.3 (minus strand)
Summary	Function	Hormone regulating blood glucose	Hydrolyses lactose to glucose + galactose
Context	Neighbours	Other human genes	Beside *lacY*, *lacA* in the *lac* operon

GenBank flat file (read in order)

1. **LOCUS**: name, length (bp), molecule type, date.
2. **DEFINITION**: one-line description.
3. **ACCESSION / VERSION**.
4. **SOURCE / ORGANISM**: species and lineage.
5. **FEATURES**: `gene`, `CDS` (start to stop codon; has `/translation` and `/protein_id`), `exon`, UTRs.
6. **ORIGIN**: the sequence.

Gene vs mRNA vs CDS. Gene = the DNA region; mRNA = processed copy; CDS = the translated part. In human mRNA the CDS is shorter than the mRNA because of the UTRs.

Protein record

Check length (110 aa for NP_000198.1), FEATURES (signal peptide, B chain, C peptide, A chain), `/db_xref`, and `/coded_by` (links back to Nucleotide). For NP_414878.1 note 1,024 aa and any listed domains/active sites.

Part 6. Comparing biological contexts

Feature	***lacZ***	***INS***
Cell type	Prokaryote	Eukaryote
Genome	One circular chromosome	23 pairs; gene on chr 11
Gene structure	Continuous, no introns	3 exons, 2 introns

Feature	*lacZ*	*INS*
Gene vs CDS	Gene ≈ CDS (3,075 bp)	Gene ≈ 1.4 kb; CDS 333 bp
mRNA record	None; annotated on genome	NM_000207.3
Arrangement	Operon (*lacZYA*), one mRNA, three proteins	Single gene, one protein
Processing	Translated directly	Preproinsulin cleaved to insulin (A+B)
Protein size	1,024 aa; tetramer	110 aa → 51 aa mature
Location label	Locus tag, coordinates	Chromosome band
Regulation	LacI repressor; lactose/IPTG	Glucose-responsive in β-cells

Takeaway: organism type decides which records exist. Bacteria give genome-based records with no NM_; humans give separate genomic, mRNA and protein records, and the mRNA is shorter than the gene because introns are removed.